NOVEMBER - DECEMBER 2025 EXAMINATION
I Yr. B.TECH. (BM, CE, CS, EE & EC Branches only)
PH10009: APPLIED PHYSICS

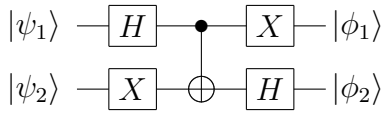
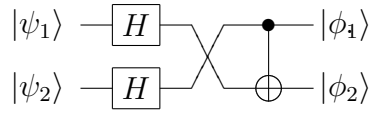
Time : 3 hours Max Marks : 70

TOTAL NUMBER OF SECTIONS IN THIS PAPER - 05.

NOTE: Missing data if any, may be assumed appropriately.

Given: $\hbar = \mathbf{1.054 \times 10^{-34}}$Js, $\boldsymbol{k_B} = \mathbf{1.38 \times 10^{-23}}$J/K, $\boldsymbol{m_e} = \mathbf{9.11 \times 10^{-31}}$kg and $\boldsymbol{c} = \mathbf{3 \times 10^{8}}$m/s.

Q. No.		QUESTION	Marks	CO	BL	PI
1	(a)	With neat sketch discuss various light - matter interaction processes.	2	1	2	1
	(b)	Match the following: (i) Population Inversion (α)Helps perform round trip of light. (ii) A Pump (β)Al_2O_3:Cr^{3+} crystal in Ruby laser. (iii) Resonator (γ)Electric discharge in He-Ne laser (iv) An Active medium (δ)$N_1 \leq N_2$	2	1	1,3	1
	(c)	Obtain an expression for the ratio of Einstein's A and B Coefficients.	3	1	4	1
	(d)	(i) What are metastable states? What is their significance in Lasing action? (ii) Why two-level lasers does not exists? (iii) Why photons are called Bosons?	3+2+2	1	4	2
		OR				
	(e)	With appropriate energy diagram and schematics, explain the working of a solid state laser and a gas laser.	3½+3½	1	1,2	3
2	(a)	Classify optical fibers based on their refractive index profile and mode profile.	2	2	1	1
	(b)	Draw a neat ray diagram of light guiding in an optical fiber. Derive an expression for maximum incident angle (i_{max}) at the input end of the optical fiber for no-loss light guiding.	2	2	3	2
	(c)	Write an essay on the contribution of any two scientists only, from the following (i) NS Kapany (ii) SN Bose (iii) CKN Patel and (iv) ECG Sudharshan.	3	2	1,6	1
	(d)	(i) A ray of light is launched at 7.5° at the air-core interface of an optical fiber of $n_{core} = 1.5$. For TIR find the minimum value $n_{cladding}$. (ii) The angle of acceptance of an optical fiber is 30° when kept in air. Find the acceptance angle when the same fiber is immersed in a liquid of refractive index 1.33 . (iii) The attenuation in optical fiber is 3.6dB/km. What fraction of its initial intensity remain after 1.5km.	3+2+2	2	4,5	3
		OR				
	(e)	(i) Draw the refractive index profiles of step-index and graded index optical fibers. (ii) If n_0, n_{co} and n_{cl}, are the refractive Indies of air, core and cladding, show that it is possible to have $NA > 1$ when $n_0 > 1$.	3+4	2	1,3	3
3	(a)	Is earth an inertial frame of reference? What about Sun? Explain.	2	3	2	1
	(b)	Find the kinetic energy (KE) in classical theory and in relativistic theory for an electron moving at a velocity of $0.99c$. Compare both values of KE and discuss.	2	3	3,4	2
	(c)	How fast would an athlete need to be running for a 100-m race to look 100-ft long? What is unreasonable? Discuss.	3	3	5	2

	(d)	Obtain expressions for the relation between mass & energy and mass & momentum. Also reduce the expressions of Energy and momentum for a massless particle.	3+3+1	3	1,3	3
		OR				
	(e)	What are Galilean and Lorentz transformation equation in x, y, z and t. Using Einstein's postulates obtain Lorentz transformations from Galilean transformations.	2+5	3	2	2,3
4	(a)	Estimate the number of photons per second, emitted from a light source of wavelength 580nm and optical power 10W.	2	4	5	2
	(b)	An atom in an excited state stores energy temporarily. If the lifetime of this excited state is measured to be 100 ps, what is the minimum uncertainty in the energy of the state?	2	4	6	2
	(c)	An electron and a photon of wavelength 500nm has same momentum. Calculate the energy of these electron and the photon and compare.	3	4	3,4	3
	(d)	(i) What is Compton effect? (ii) Obtain an expression for the wavelength of scattered photon in Compton effect. (iii) If incident X-rays have a wavelength of 10.0 pm, the scattered X-rays at a certain angle have a wavelength of 10.5 pm. Find the momentum (magnitude and direction) of the corresponding recoil electrons.	1+3+3	4	1,3	1,2
		OR				
	(e)	(i) What are the significance of "Normalization Condition"? Explain. If $\psi = Ae^{-i(Et/\hbar + i\alpha x/2)}$ is a normalized wave function, find the value of A. (ii) Using Schrödinger's equation, obtain a normalised wavefunction for a particle in a one dimensional potential well. Also plot the probability amplitude and probability for $n = 1, 2, 3$.	2+5	4	2,4	1,3
5	(a)	What are unitary operations? Does, any classical Gate has unitary property?	2	5	6	1,2
	(b)	What do you understand by the word "entanglement"?	2	5	2	1
	(c)	In the following cases find the $-\boxed{?}-$, (i) $\|0\rangle - \boxed{H} - \boxed{?} - \|0\rangle$ (ii) $\|1\rangle - \boxed{?} - \boxed{Z} - \|1\rangle$	3	5	6	2
	(d)	Solve the following:  for all possible 2-Qubit inputs ($\|\psi_1\psi_2\rangle$) and find output ($\|\phi_1\phi_2\rangle$)for each case. Report the results in the form of Truth-Table.	7	5	5	3
		OR				
	(e)	Find the output for all possible inputs ($\|\psi_1\psi_2\rangle$) :  Explain the output qubits in each case.	7	5	4	2,3

CO-Course outcome. PI-Performance Indicator (1.Cognitive, 2. Affective, 3.Psychomotor). BL-Bloom's Taxonomy Levels: 1-Remember, 2-Understand, 3-Apply, 4-Analyze, 5-Evaluate, 6-Create.

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ANSWER KEYS and RUBRICS FOR Q. No. 1 — Max Marks: 14

Q.	No.	QUESTION & ANSWER KEY	Marks
1	(a)	With neat sketch discuss various light - matter interaction processes.	2
		Absorption, spontaneous emission and stimulated emission process mentioned clearly	½
		Sketch of all three processes given and explained	3×½
	(b)	Match the following: (i) Population Inversion (δ) $N_1 \leq N_2$ (ii) A Pump (γ) Electric discharge in He-Ne laser (iii) Resonator (α) Helps perform round trip of light. (iv) An Active medium (β) Al_2O_3:Cr^{3+} crystal in Ruby laser.	2
		If all answers matched as above.	4×½
	(c)	Obtain an expression for the ratio of Einstein's A and B Coefficients.	3
		Brief about all radiation-matter interaction processes	1
		Obtained rate equation and the ratio (N_1/N_2)	1
		Obtained the expression for A_{21}/B_{21} and discussed.	1
	(d)	(i) What are metastable states? What is their significance in Lasing action? (ii) Why two-level lasers does not exists? (iii) Why photons are called Bosons?	3+2+2
		(i) Energy diagram given for metastable state.	1½
		Longer lifetime explained	1½
		(ii)Under steady pumping, rate of absorption and rate of emission never exceed 1 or $N_2 > N_1$ is impossible	1
		Addition of a metastable state helps overcome this.	1
		(iii) S N Bose and contributions described. Name of Boson justified.	2
	(e)	With appropriate energy diagram and schematics, explain the working of a solid state laser and a gas laser.	3½+3½
		Energy diagram given correctly, with correct marking of lasing lines	1
		All components of lasers shown correctly	1
		Working of laser explained	1
		Population inversion and metastable states mentioned.	½

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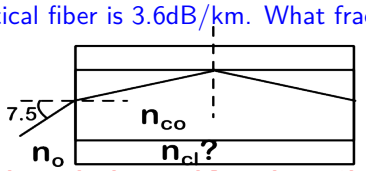
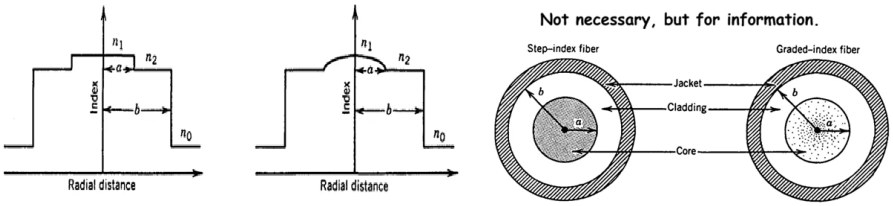
ANSWER KEYS and RUBRICS FOR Q. No. 2 — Max Marks: 14

Q.	No.	QUESTION & ANSWER KEY	Marks
2	(a)	Classify optical fibers based on their refractive index profile and mode profile.	2
		SIF & GIF mentioned against index based classification.	1
		SMF & MMF mentioned againet mode based classification	1
	(b)	Draw a neat ray diagram of light guiding in an optical fiber. Derive an expression for maximum incident angle (i_{max}) at the input end of the optical fiber for no-loss light guiding.	2
		Ray diagram which has all rays, refractive indies mentioned	1
		correct derivation for $i_{max} = \sin^{-1}(\sqrt{n_2 - n_1})$	1
	(c)	Write an essay on the contribution of any **two scientists only**, from the following (i) NS Kapany (ii) SN Bose (iii) CKN Patel and (iv) ECG Sudharshan.	3
		Contribution of Scientist 1	1½
		Contribution of Scientist 2	1½
	(d)	(i) A ray of light is launched at 7.5° at the air-core interface of an optical fiber of $n_{core} = 1.5$. For TIR find the minimum value $n_{cladding}$. (ii) The angle of acceptance of an optical fiber is 30° when kept in air. Find the acceptance angle when the same fiber is immersed in a liquid of refractive index 1.33 . (iii) The attenuation in optical fiber is 3.6dB/km. What fraction of its initial intensity remain after 1.5km.	3+2+2
			1
		(i) Ray diagram as above made and the problem described correctly.	
		$\sin\theta_c = 0.087$, $\theta_c = 5°$	1
		Expression for n_{clad} obtained and obtained $n_{clad} \leq 1.49$.	1
		(ii)Expression for $NA = n_0 \sin\theta_a$ is given correctly.	1
		For $n_o = 1.33$, $\theta_a = 22°$	1
		(iii)Total attenuation A is 3.6dB/km × 1.5km = 5.4dB.	1
		The expression $I = I_0 10^{-A/10}$is given correctly and $I/I_0 = 0.29$ or 29%.	1
	(e)	(i) Draw the refractive index profiles of step-index and graded index optical fibers. (ii) If n_0, n_{co} and n_{cl}, are the refractive Indies of air, core and cladding, show that it is possible to have $NA > 1$ when $n_0 > 1$.	4+3
			
		(i) Refractive index profile of each fiber drawn correctly	1+1
		The x-axis marked with core/cladding rdius and y-axis marked with n_c, n_{cl} and n_0	1+1
		Since $\sin\theta_i = \sqrt{n_{co}^2 - n_{cl}^2}$ this can not have values > 1.	1
		Also, $NA = n_o\sqrt{n_{co}^2 - n_{cl}^2}$	1
		Thus, in an immersion medium with $n_0 > 1$, a fiber (or any optical system) can indeed have $NA > 1$.	1

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ANSWER KEYS and RUBRICS FOR Q. No. 3 — Max Marks: 14

Q.	No.	QUESTION & ANSWER KEY	Marks
3	(a)	Is earth an inertial frame of reference? What about Sun? Explain.	2
		Yes and No. For an observer making measurements which are at small ranges, the earth can be treated as an inertial frame, for a large measurements high speed/velocity at long ranges earth is not an IFR.	1
		For solar system sun is assumed to be heliocentric, hence it can be treated as IFR, however for measurements involving inter galactic, for other solar systems, sun would be Non-IFR.	1
	(b)	Find the kinetic energy (KE) in classical theory and in relativistic theory for an electron moving at a velocity of 0.99c. Compare both values of KE and discuss.	2
		Classical KE= $\frac{1}{2}mv^2 \approx 4 \times 10^{-14}$J and Relativistic KE= $\frac{m_o c^2}{\sqrt{1-\frac{v^2}{c^2}}} \approx 50$J.	1
		The classical formula underestimates the actual energy because it does not account for the increase in relativistic mass and the full energy-momentum relation. At high velocities (close to c), relativistic effects become dominant, and classical mechanics fails to give accurate results.	1
	(c)	How fast would an athlete need to be running for a 100-m race to look 100-ft long? What is unreasonable? Discuss.	3
		Relativistic length contraction is $L = L_0\sqrt{1-\frac{v^2}{c^2}}$	1
		$\frac{L}{L_0} \approx 30/100 = 0.33 \Rightarrow v = 0.95c$	1
		so the required speed is extremely close to c, which is physically impossible for an athlete.	1
	(d)	Obtain expressions for the relation between mass & energy and mass & momentum. Also reduce the expressions of Energy and momentum for a massless particle.	3+3+1
		Starting with Newton's second law obtained the relation $E = \frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}}$	2
		Mentioned rest energy $E_0 = mc^2$	1
		Obtained the expression $E = \sqrt{(mc^2)^2 + p^2c^2}$	3
		For $m = 0$, $E = pc$ in all frame of reference.	1
	(e)	What are Galilean and Lorentz transformation equation in x, y, z and t. Using Einstein's postulates obtain Lorentz transformations from Galilean transformations.	2+5
		Galilean transformation equations $x' = x - vt$, $y' = y$, $z' = z$, and $t' = t$	1
		Lorentzian transformation equations $x' = \gamma(x - vt)$, $y' = y$, $z' = z$, and $t' = \gamma(t - \frac{vx}{c^2})$	1
		Given Einstein's both postulates	1
		Starts with assuming $x' = k(x - vt)$	1
		Obtained the Lorentz transformation equations correctly.	3

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ANSWER KEYS and RUBRICS FOR Q. No. 4 — Max Marks: 14

Q.	No.	QUESTION & ANSWER KEY	Marks
4	(a)	Estimate the number of photons per second, emitted from a light source of wavelength 580nm and optical power 10W.	2
		Optical power = Total Energy per second = 10W=10J/s	½
		Energy per photon = $h\nu = hc/\lambda$ J	½
		Number of photons/s = Total Energy/Energy per photon/s $\approx 2.9 \times 10^{19}$photons/s	1
	(b)	An atom in an excited state stores energy temporarily. If the lifetime of this excited state is measured to be 100 ps, what is the minimum uncertainty in the energy of the state?	2
		$\Delta E \Delta t \geq \hbar/2$ or $h/4\pi$	1
		$\Delta E \approx 5.3 \times 10^{-25}$J or $\Delta E \approx 3.3 \times 10^{-6}$eV	1
	(c)	An electron and a photon of wavelength 500nm has same momentum. Calculate the energy of these electron and the photon and compare.	3
		$E_{photon} = hc/\lambda \approx 3.98 \times 10^{-19}$J	1
		$E = \sqrt{p^2c^2 + (mc^2)} = \sqrt{(hc/\lambda)^2 + (mc^2)} \approx 8.2 \times 10^{-14}$J	1
		The electron's energy is much greater than the photon's, because most of the electron's energy is its rest mass energy, while the photon's energy is entirely kinetic.	1
	(d)	(i) What is Compton effect? (ii) Obtain an expression for the wavelength of scattered photon in Compton effect. (iii) If incident X-rays have a wavelength of 10.0 pm, the scattered X-rays at a certain angle have a wavelength of 10.5 pm. Find the momentum (magnitude and direction) of the corresponding recoil electrons.	1+3+3
		Compton effect: Scattering of electron by a photon, which inturn increase wavelength (decrease in energy) of photons, demonstrating the particle-like behavior of light and providing evidence for quantum mechanics.	1
		Neat sketch of compton effect.	1
		Derived the expression for $\lambda' - \lambda = \lambda_c(1 - \cos\phi)$, correctly.	2
		Let $p_i = h/\lambda$, $p_s = h/\lambda'$ and $p_e = \sqrt{p_i^2 + p_s^2 - 2p_ip_s\cos\phi}$ are the momentum of incident photon, scattered photon and scattered electron, respectively.	
		Obtained $\cos\phi = 0.793$ and $\phi \approx 37.5°$	1
		$p_e = 1.3 \times 10^{-22}$kg.m/s	1
		Scattering angle for electron is $\tan\theta = \frac{p_s \sin\phi}{p_i - p_s\cos\theta}$, $\theta = 67°$	1
	(e)	(i) What are the significance of "Normalization Condition"? Explain. If $\psi = Ae^{-i(Et/\hbar + i\alpha x/2)}$ is a normalized wave function, find the value of A. (ii) Using Schrödinger's equation, obtain a normalised wavefunction for a particle in a one dimensional potential well. Also plot the probability amplitude and probability for $n = 1, 2, 3$.	2+5
		(i) The normalization condition in quantum mechanics ensures that the total probability of finding a particle somewhere in space is exactly 1. This is essential because the wave function $\psi(x,t)$ represents the probability amplitude, and its square modulus $\lvert\psi(x,t)\rvert^2$ gives the probability density.	1
		In a region between $x = 0$ and L, ψ the normalization yields the constant as $A = \sqrt{1/\alpha}$	1
		(ii) One dimensional potential well sketched and explained boundaries, and 1d Schrödinger's given.	1
		Derived equation using Schrödinger's equation and normalised	1+1
		$\psi(x)$ and $\lvert\psi(x)\rvert^2$ plotted for $n = 1, 2, 3$.	1+1

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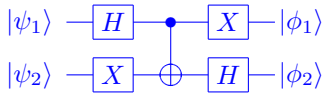
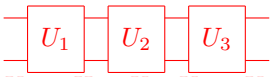
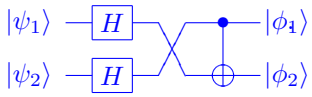
ANSWER KEYS and RUBRICS FOR Q. No. 5 — Max Marks: 14

Q.	No.	QUESTION & ANSWER KEY	Marks
5	(a)	What are unitary operations? Does, any classical Gate has unitary property?	2
		Unitary operations are transformations in quantum mechanics that preserve the inner product and the norm of quantum states, ensuring that probabilities remain conserved. Mathematically, a unitary operator U satisfies $U^\dagger U = UU^\dagger = I$, where $U^\dagger$ is the adjoint of U.	1
		Classical gates does not have unitary operation, but NOT gate exhibits single qubit unitary like behaviour.	1
	(b)	What do you understand by the word "entanglement"?	2
		Entanglement is a quantum phenomenon where the states of two or more particles become interconnected in such a way that the state of one particle cannot be described independently of the others, even when separated by large distances.	1
		The correlation strong such that measuring one entangled particle instantly determines the state of its partner, regardless of the distance between them.	1
	(c)	In the following cases find the $-\boxed{?}-$, (i) $\|0\rangle - \boxed{H} - \boxed{?} - \|0\rangle$	3
		(ii) $\|1\rangle - \boxed{?} - \boxed{Z} - \|1\rangle$	
		For (i) the correct answer is $-\boxed{H}-$	
		For (ii) the correct answer is $-\boxed{Z}-$	
		Matrix form or logic of Z and H gates defined	½ each
		Correct answer solved using matrix, or using logic with proof.	1 each
		Correct answer given without any proof.	½ each
	(d)	Solve the following:	7
			
		for all possible 2-Qubit inputs ($\|\psi_1\psi_2\rangle$) and find output ($\|\phi_1\phi_2\rangle$)for each case. Report the results in the form of Truth-Table.	
		 $\Rightarrow \|\psi\rangle - \boxed{U} - \|\phi\rangle$ where $U = U_3U_2U_1$	1+1
		$U_1 = H \otimes X$, $U_2 = U_{CNOT}$ and $U_3 = X \otimes H$ and obtained correctly	3
		Truth table for $\|00\rangle, \|01\rangle, \|10\rangle, \|11\rangle$ and the corresponding outputs given in table format. See the table shown below for (5d).	2
	(e)	Find the output for all possible inputs ($\|\psi_1\psi_2\rangle$) :	7
			
		Explain the output qubits in each case.	
		$U_1 = H \otimes H$, $U_2 = U_{SWAP}$ and $U_3 = U_{CNOT}$ defined in matrix format.	1+1+1
		Correct matrix for $U = U_3U_2U_1$ obtained	2
		Truth table for $\|00\rangle, \|01\rangle, \|10\rangle, \|11\rangle$ and the corresponding outputs given in table format. See the table shown below for (5e).	2

Truth table for (5d)

$\|\psi_1\psi_2\rangle$	$\|\phi_1\phi_2\rangle$
$\|00\rangle$	$(\|00\rangle + \|01\rangle + \|10\rangle - \|11\rangle)/2$
$\|01\rangle$	$(\|00\rangle - \|01\rangle + \|10\rangle + \|11\rangle)/2$
$\|10\rangle$	$(-\|00\rangle - \|01\rangle + \|10\rangle - \|11\rangle)/2$
$\|11\rangle$	$(-\|00\rangle + \|01\rangle + \|10\rangle + \|11\rangle)/2$

Truth table for (5e)

$\|\psi_1\psi_2\rangle$	$\|\phi_1\phi_2\rangle$
$\|00\rangle$	$(\|00\rangle + \|01\rangle + \|10\rangle + \|11\rangle)/2$
$\|01\rangle$	$(\|00\rangle - \|01\rangle - \|10\rangle + \|11\rangle)/2$
$\|10\rangle$	$(\|00\rangle - \|01\rangle + \|10\rangle - \|11\rangle)/2$
$\|11\rangle$	$(\|00\rangle + \|01\rangle - \|10\rangle - \|11\rangle)/2$

Attention of Students

Challenge to answer keys,
may please be submitted
on or before Dec. 30, 2025,
in person at Dept. Appl. Physics
OR
scanned copy may be mailed to
jtandrews@sgsits.ac.in

Please, submit: Q. No., alternate or correct solution/answer